

UNIT 1 – ELECTRIC CHARGE, FORCE, AND FIELD

Unit 1 Review**Answer key**

Placeholder worksheet for the site prototype. Free for classroom use.

1. Two point charges, $q_1 = +3.0 \mu\text{C}$ and $q_2 = -5.0 \mu\text{C}$, are 0.20 m apart. Find the magnitude of the force each exerts on the other and state whether it is attractive or repulsive.

$$F = \frac{k|q_1q_2|}{r^2} = \frac{(8.99 \times 10^9)(3.0 \times 10^{-6})(5.0 \times 10^{-6})}{(0.20)^2} = 3.4 \text{ N. Opposite signs, so the force is attractive.}$$

2. Find the magnitude of the electric field 0.050 m from a point charge of +8.0 nC.

$$E = \frac{kq}{r^2} = \frac{(8.99 \times 10^9)(8.0 \times 10^{-9})}{(0.050)^2} = 2.9 \times 10^4 \text{ N/C, directed away from the charge.}$$

3. An electron enters a region of uniform field $E = 500 \text{ N/C}$ with speed $2.0 \times 10^6 \text{ m/s}$ perpendicular to the field. How far has it been deflected sideways after travelling 4.0 cm along its original direction? Ignore gravity.

$$\begin{aligned} \text{Time in the field: } t &= \frac{0.040}{2.0 \times 10^6} = 2.0 \times 10^{-8} \text{ s. Acceleration: } a = \frac{eE}{m_e} = 8.8 \times 10^{13} \text{ m/s}^2. \\ \text{Deflection: } y &= \frac{1}{2}at^2 = \frac{1}{2}(8.8 \times 10^{13})(2.0 \times 10^{-8})^2 = 1.8 \times 10^{-2} \text{ m} = 1.8 \text{ cm.} \end{aligned}$$